The Entangled Mind: Quantum Cognition as a Cognitive Architecture for Language Models

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Author Note

Sean Pharris is an independent researcher with a background in psychology and philosophy and professional experience in prompt engineering and AI pipeline development. This document is a pre-registration-style research proposal; no data have been collected.

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Abstract

Do machines need quantum minds? Every language model built to date reasons under classical probability: its beliefs are separate quantities that are weighed and added. Human judgment does not work this way. Three decades of findings in psychology (conjunction effects, order effects, context dependence) are better described by the mathematics of quantum probability, in which beliefs can be in superposition, can interfere, and, crucially, can be entangled, so that measuring one belief changes another that was never touched. Quantum cognition has modeled these effects in people but has never been built into a machine. Quantum machine learning has built circuits but has never asked a cognitive question. This proposal joins the two. We extract the competing contentions from a language model's hidden reasoning trace, encode them as an entangled state on real quantum hardware, let that state evolve under the structure of their mutual support and contradiction, and read the result as the model's judgment. The primary question is not whether this is faster or more accurate. It is whether a mind whose beliefs are entangled judges differently from one whose beliefs are merely separable, and whether anything appears in that difference that neither the input thoughts nor the classical baseline contained. We pre-register falsifiable tests for each claim, including a shuffled-structure control that would falsify the idea if the quantum encoding carries no information. The work is framed deliberately at the intersection of physics, psychology, philosophy, and computer science, because the question it asks, what kind of thing a judging mind is, belongs to all of them.

Keywords: quantum cognition, entanglement, language models, cognitive architecture, contextuality, emergence, judgment

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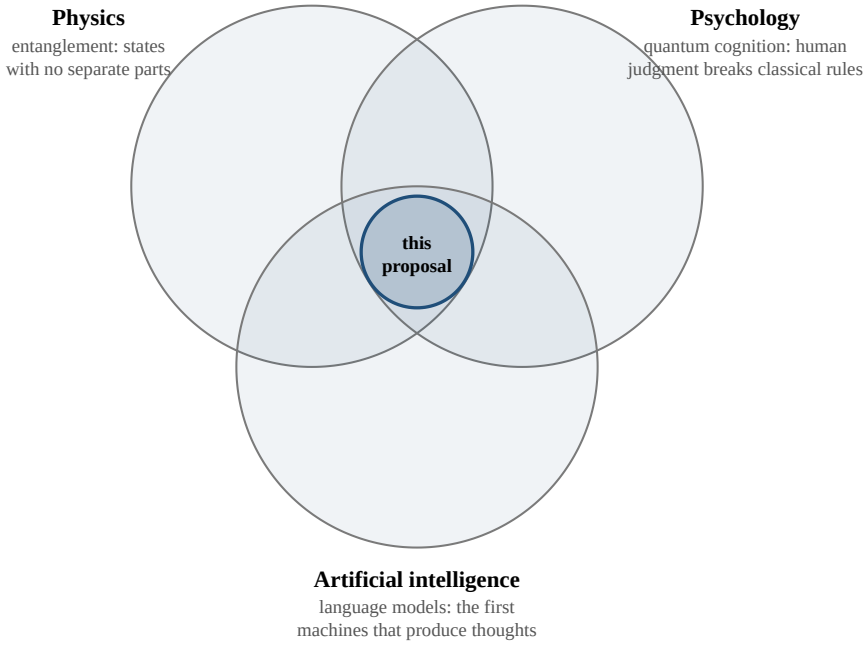
This proposal describes an experiment that has not been run. It sets out the idea, the architecture that would test it, the hypotheses, the controls that could falsify each one, and the criteria by which the result will be judged before any data exist. It is written for readers in several fields at once, so the early sections stay above the trees; the method and analysis sections go into them.

Why This Matters

Three fields are circling the same question and not speaking to each other. Physics has a mathematics of states that cannot be decomposed into parts: entanglement. Psychology has thirty years of evidence that human judgment violates the rules of classical probability in exactly the ways that mathematics predicts (Busemeyer & Bruza, 2012; Pothos & Busemeyer, 2013). Artificial intelligence has built the first machines that produce something like thoughts, and has built every one of them on classical probability. Each field holds a piece. None has put the pieces in one room (Figure 1).

Figure 1

Three Fields Hold the Pieces of One Question



Note. The overlap is where this proposal sits: a machine whose competing beliefs form a single entangled state. Physics supplies the mathematics, psychology supplies the evidence that human judgment already behaves this way, and language models supply the beliefs.

The hard problem of consciousness (Chalmers, 1995) is not a side note here. We do not claim to explain consciousness. But the hard problem has one feature that this work touches directly: a conscious mind seems to be a unity, not a committee. Its beliefs are not a list; they are a single state that changes as a whole. Classical systems can only approximate that unity by adding up parts. Entangled states are, mathematically, the thing itself: a whole that has no description in terms of its parts. Whether a machine built on such states behaves differently is a question that can be run, measured, and published. That moves a corner of the hard problem from philosophy into experiment.

Current systems are not the only possible shape for a thinking machine. Transformers (Vaswani et al., 2017) were the first architecture to work at scale, and the field has, understandably, built everything on top of them. But the first thing to work is rarely the final

form. If the judgment step of a language model can be lifted out and rebuilt on a different mathematical foundation, then the architecture of thinking machines is still an open design space, and this paper is an argument that the space is worth exploring.

If entangled beliefs produce judgments that are measurably different, measurably more human-like, or contain structure that the inputs did not, then three things follow. Scientifically, quantum cognition gains its first working model instead of a descriptive fit. Philosophically, the question of what a mind is gains an experimental handle. Practically, a new layer becomes possible in every AI system: one that does not average conflicting thoughts but lets them cancel, and that can be watched while it does so.

A clean negative result is still a contribution. It would tell the field, with data, that the quantum structure of human judgment is a description and not a mechanism, and that machine minds can safely stay classical. Either answer is worth having. The point is that nobody has asked.

Background

Quantum Cognition

Beginning around 2009, Busemeyer, Bruza, Pothos, Aerts, and colleagues showed that several stubborn anomalies in human judgment are naturally predicted by quantum probability (Aerts, 2009; Busemeyer & Bruza, 2012; Bruza et al., 2015). The conjunction fallacy, in which participants judge “Linda is a feminist bank teller” more likely than “Linda is a bank teller” (Tversky & Kahneman, 1983), is modeled as interference between incompatible framings. Order effects, where the same two questions yield different answers in different orders, are modeled as non-commuting measurements (Wang et al., 2014). Context effects, where deciding one thing changes a belief about an unrelated thing, are modeled by entangled belief states. These models fit human data better than classical ones, and they are explicitly agnostic about whether the brain is physically quantum. They are models of the mathematics of judgment, not of neurons. No one has run them on a machine that judges.

Quantum Machine Learning

Google, IBM, Xanadu, and others have built variational quantum circuits for classification and optimization, largely as a bet on future fault-tolerant hardware (Biamonte et al., 2017). Current devices offer on the order of a thousand noisy qubits; free public tiers offer tens. The field asks whether quantum circuits can learn faster or represent more. It has not asked whether a quantum state can hold a belief.

Language Models and Hidden Reasoning

Frontier language models now produce extended reasoning traces before answering. These traces are sequential self-talk under classical probability: a contradiction raised early can be outvoted by later tokens but cannot cancel against them. Interpretability research examines the wiring (which internal features activate for which concept) rather than the dynamics (how competing beliefs resolve). The trace is, however, a rich and inexpensive source of a model's competing contentions, which makes it the natural input to a system that wants to hold those contentions in one state.

What Is Missing

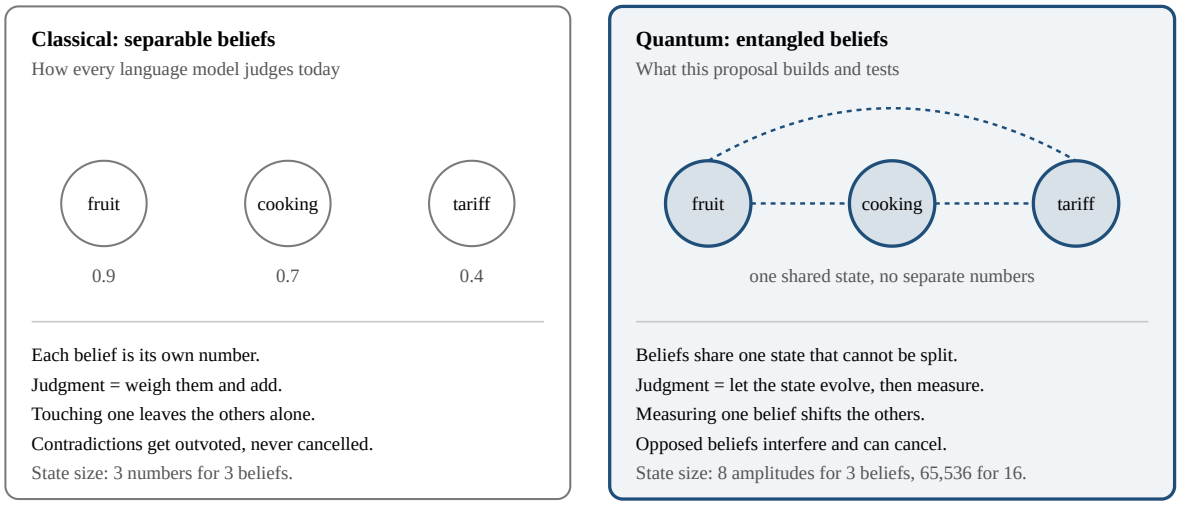
A system in which a language model's competing contentions are encoded as a single entangled quantum state, allowed to evolve under their mutual relations, measured as a judgment, and watched while it happens. Each ingredient exists. The combination does not.

The Central Idea

Consider a model answering “Is a tomato a vegetable?” Its reasoning trace contains several contentions: botanically a fruit; used as a vegetable in cooking; ruled a vegetable by a court for tariff purposes; the question is probably about cooking. Today's models hold these as separate quantities and combine them. The proposal holds them as one quantum state (Figure 2).

Figure 2

Separable Beliefs Are Added; Entangled Beliefs Are One State



Note. Left: the classical picture every current language model uses. Right: the entangled picture this proposal builds. Dashed links indicate entanglement. Numbers in the left panel are illustrative confidence weights. The right panel is the only place emergence could live; the left panel is what it is compared against.

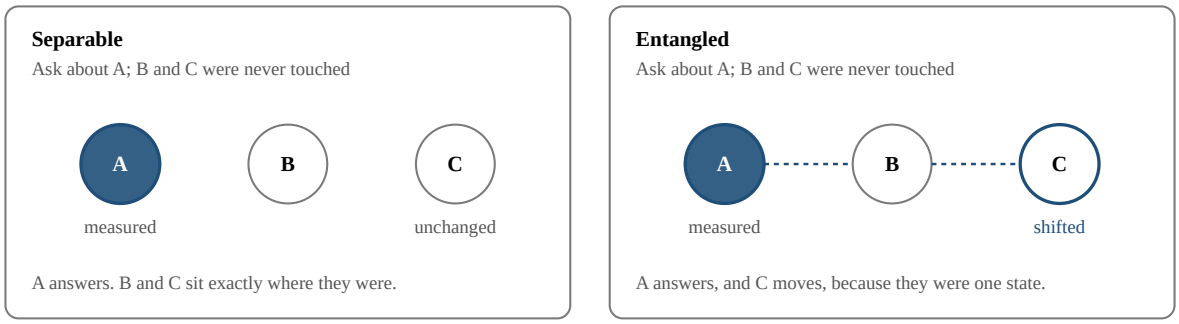
The difference is not cosmetic. In the separable picture, three beliefs are three numbers, and the system's whole state is just the list. In the entangled picture, three beliefs are a single state with eight amplitudes, and sixteen beliefs are a single state with 65,536. The state is a pattern across all of them that no single belief owns. Three consequences follow, and each is a testable claim.

Interference. Beliefs that oppose each other do not merely outweigh one another; they can cancel, leaving the system with less confidence than either belief alone. Beliefs that support each other can reinforce beyond their sum.

Contextuality. Measuring one belief (asking about tariffs) changes the state of beliefs that were never addressed (cooking). This is the signature of entanglement, and it is precisely the human effect that quantum cognition was invented to model (Figure 3).

Figure 3

Contextuality: Measuring One Belief Moves an Untouched One Only When They Share a State



Note. In both panels only belief A is measured. In the separable case B and C are unaffected; in the entangled case C shifts in the direction the relation web predicts. Hypothesis H3 tests for the right-hand behavior on real hardware.

Evolution before collapse. The entangled state can be allowed to drift under an energy landscape built from the beliefs' mutual relations before any question is asked. What settles out may not be any of the inputs. This is where the proposal looks for emergence: structure in the judgment that neither the contentions nor a classical combination of them contained.

A bare one-qubit-per-belief design with no entanglement is a smoke test of the pipeline and nothing more; a laptop simulates it exactly, and nothing new can appear in it. Entanglement is the whole bet.

Hypotheses

Each hypothesis is paired with the null result that would falsify it (Table 1). H2 is the pre-registered primary; the others are secondary and exploratory. Figure 4 shows how they depend on one another.

Table 1

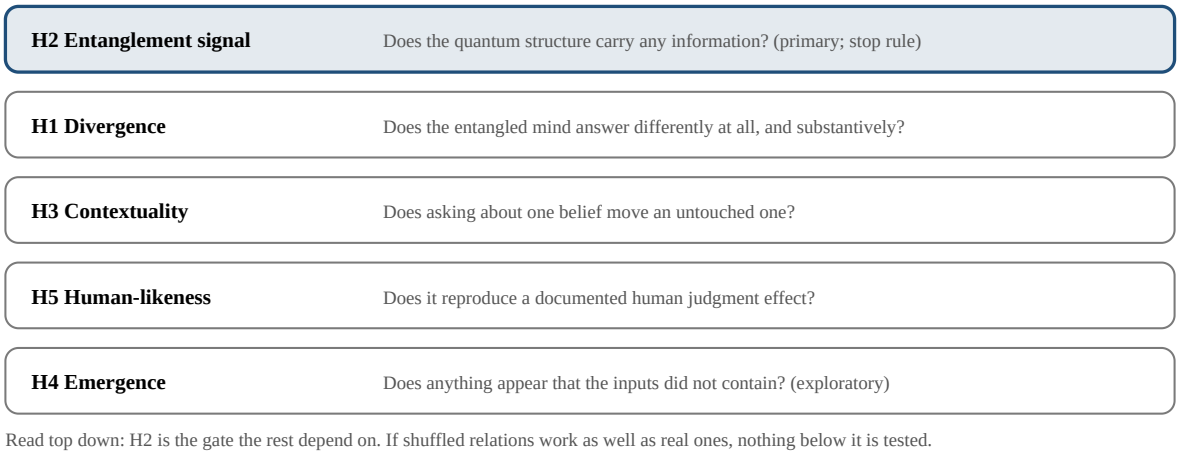
Hypotheses and the Null Result That Would Falsify Each

Label	Hypothesis	Null (what falsifies it)
H1 Divergence	An entangled judgment layer produces answers that differ from the base model on a meaningful fraction of questions, and a blind rater finds the disagreements substantive rather than noise.	Agreement above 95%, or disagreements rated as noise.
H2 Entanglement signal (primary)	The entangled encoding flags conflict on known-wrong answers at a rate at least double that of (a) the same pipeline with no entangling gates and (b) the same pipeline with shuffled relations.	Either control performs as well. The quantum structure then carries no information.
H3 Contextuality	Measuring the state on one contention measurably shifts the readout of an unrelated contention, in the direction the relation web predicts.	No shift beyond hardware noise, or a shift in a random direction.
H4 Emergence (exploratory)	After evolution, the judgment contains structure (a conclusion, a flagged tension, a ranking) present in neither the input contentions nor the classical baseline.	Everything in the output is traceable to a single input or to their weighted sum.
H5 Human-likeness	The layer reproduces at least one documented human judgment effect (conjunction, order, or context) with the same sign as human data.	Wrong sign or no effect.

Note. Pass criteria for each hypothesis are given in the Analysis Plan.

Figure 4

The Hypotheses Form a Ladder With H2 as the Gate



Note. H2 is tested first because it decides whether the quantum encoding carries any information at all. H4 sits last because it is the hardest to pre-register honestly; it is given a procedure rather than a threshold.

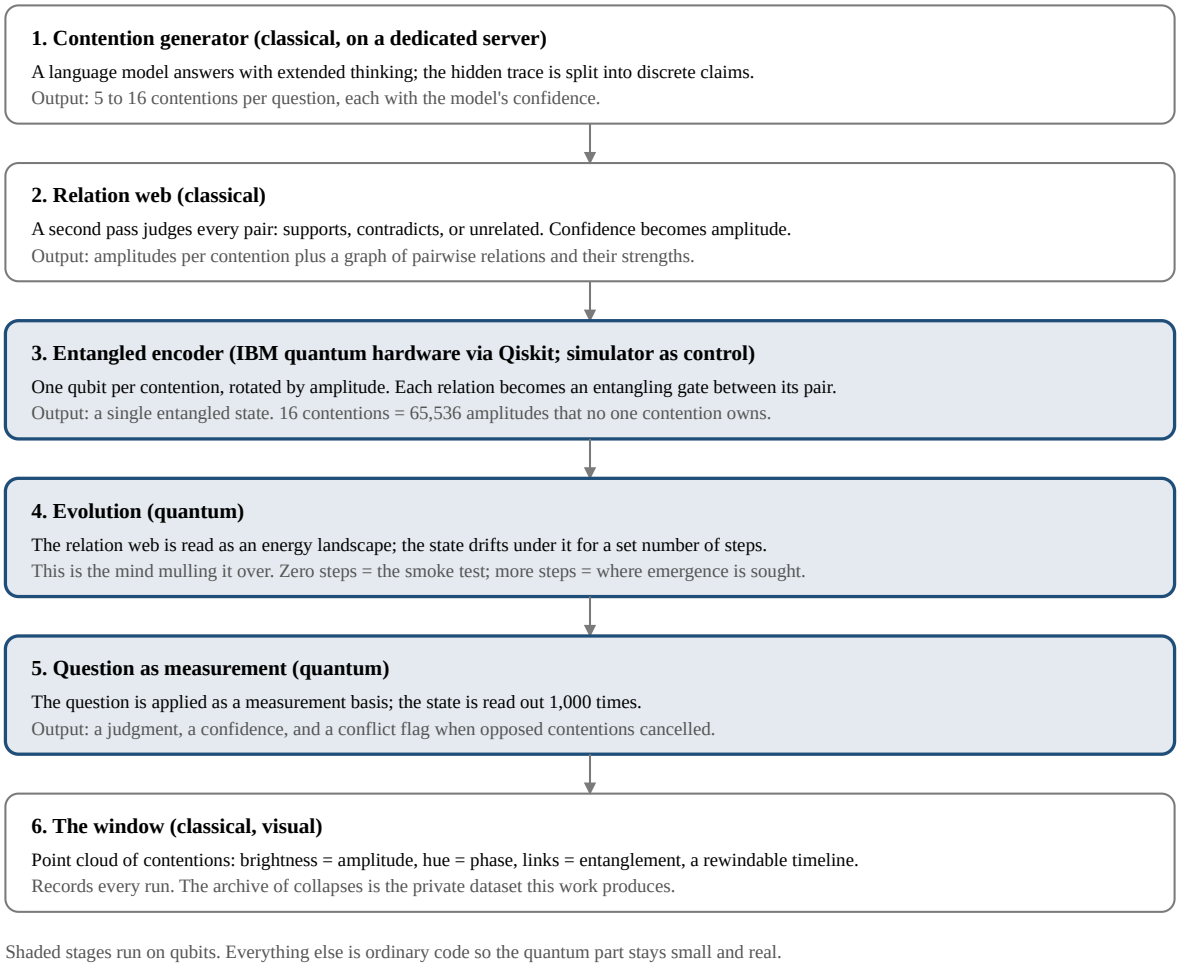
H4 is deliberately exploratory. It is the hypothesis this work most hopes to say something about and the one for which a pre-registered criterion is hardest to write honestly. The analysis plan gives it a procedure rather than a threshold.

Proposed Architecture

The system is a pipeline of six stages (Figure 5). Three run on real quantum hardware; the rest are ordinary code, kept classical on purpose so the quantum part stays small enough to run today and honest enough to compare against a simulator.

Figure 5

Proposed Architecture: Six Stages, Three on Quantum Hardware



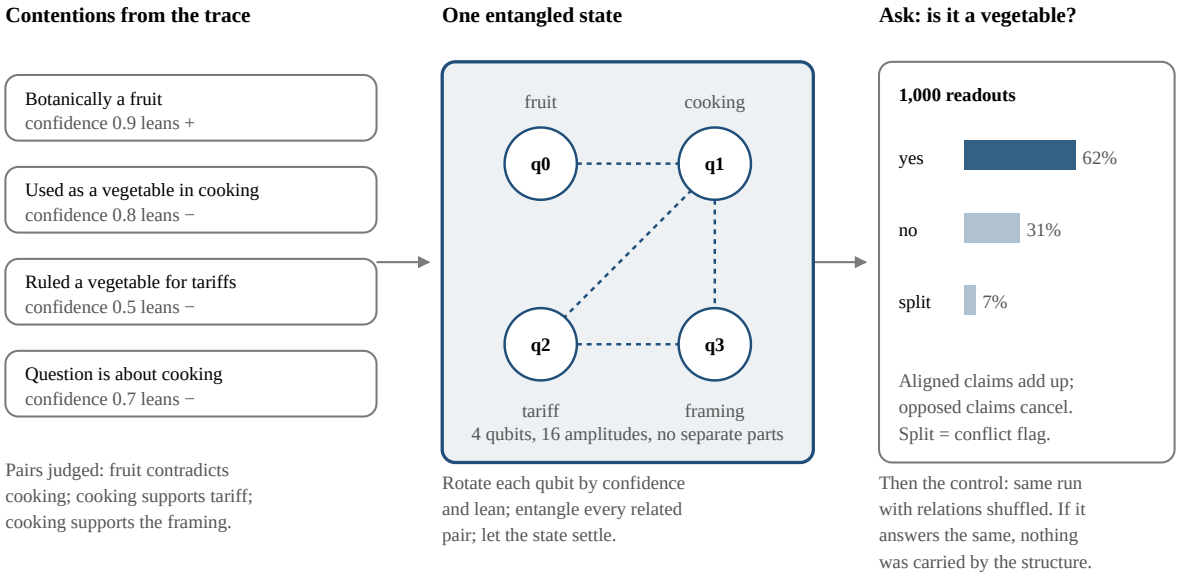
Note. Stages 3 through 5 execute on IBM Quantum devices through Qiskit (Javadi-Abhari et al., 2024), with the Qiskit Aer simulator as a control arm. Stages 1, 2, and 6 are classical and run on a dedicated server.

How a claim becomes a qubit. A contention's amplitude is the model's confidence in it, scaled to a rotation angle. Its phase is not assigned alone: the relation web is solved for a set of angles such that supporting contentions sit near each other and contradicting ones sit about 180 degrees apart. Each contention's qubit is rotated by its amplitude and phase. Then, and this is the step the earlier smoke-test design lacked, each relation in the web becomes an entangling gate between its two qubits, with a strength set by the relation's strength. After this step there are no

individual contentions left in the state; there is one state. Figure 6 walks the tomato example through it.

Figure 6

A Worked Example: Four Contentions Become One State and One Readout



Note. The contentions, confidences, and readout percentages shown are illustrative. The shuffled-relations control (right column, lower text) is the comparison that decides whether the structure carried anything.

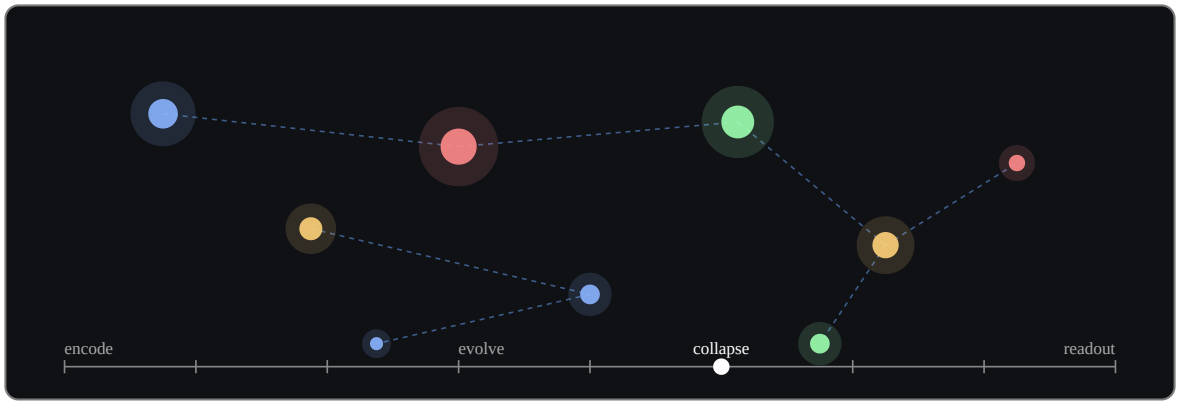
How the state evolves. The relation web is read as an energy landscape: supporting pairs lower the energy when aligned, contradicting pairs lower it when opposed. The state is evolved under this landscape for a chosen number of steps before measurement. With zero steps the pipeline reduces to the smoke test. With more steps, the state settles toward configurations that the inputs did not explicitly contain, and this is the regime in which H4 is examined.

How the question becomes a measurement. The question is encoded as a measurement basis oriented toward its “yes” answer. The state is read out 1,000 times. Contentions aligned with the answer accumulate counts; opposed ones cancel. The counts give a judgment and a confidence, and a readout split between strong opposed configurations is reported as a conflict flag.

The window. Every run is recorded and rendered as a point cloud: each contention a glowing node, brightness its amplitude, hue its phase, dashed links its entanglement, with a timeline that can be scrubbed backward through evolution and collapse (Figure 7). The window exists for two reasons. It is the only way a human can see what the state is doing, and the archive it produces is the dataset nobody else has.

Figure 7

The Window: What an Observer Sees During One Run



Each glowing point is one contention. Brightness is amplitude (how strongly it is held); colour is phase (which way it leans); dashed links are entanglement. The scrubber at the bottom rewinds through evolution to the moment of collapse. Every run is recorded. The archive is the dataset nobody else has.

Note. A sketch of the intended display, not output from a run. The scrubber rewinds through encoding, evolution, and collapse; every run is archived.

Method

Materials

A frontier language model with extended thinking exposed, run from a dedicated server. IBM Quantum via Qiskit, free tier, 5 to 16 qubits, with the Qiskit Aer simulator as the control arm and for sets larger than the hardware allows. A question set of 200 items with known answers, half chosen because the base model is known to answer them incorrectly, drawn from existing reasoning benchmarks. A belief pool of approximately 3,000 neutral prompts used to calibrate the extractor and relation judge before any test question is run.

Procedure

For each question: (a) obtain the model's reasoning trace and answer; (b) split the trace into discrete contentions with confidence, hand-checking the first 50 for extraction error; (c) run the relation judge over all pairs; (d) build and run four arms of the circuit, each with 1,000 shots: the entangled arm on hardware, the entangled arm on simulator, a no-entanglement arm, and a shuffled-relations arm; (e) record judgment, confidence, and conflict flag per arm; (f) archive the full state trajectory to the window.

Contextuality probe (H3). For a subset of 50 questions, measure one contention alone, then re-measure the full state, and compare the readout of a contention the relation web marks as linked against one it marks as unrelated.

Human-effect probe (H5). A small set of classic quantum-cognition items (the Linda problem; order-reversed question pairs) is run through the layer and compared against published human response patterns (Tversky & Kahneman, 1983; Wang et al., 2014).

Measures

Agreement rate with the base model; conflict-flag rate on known-wrong versus known-right items, per arm; blind ratings of 30 disagreements by a rater who does not know which arm produced them; readout shift under the contextuality probe; sign of effect on the human-effect items; and hardware-versus-simulator divergence as a noise estimate.

Analysis Plan and Pass Criteria

These criteria are fixed before data collection. They are open to argument now and locked afterward.

H2 (primary). Passes if the entangled arm flags conflict on at least 10% of known-wrong items, and both the no-entanglement arm and the shuffled-relations arm flag at no more than half that rate. The shuffled control is the one that matters: if random relations work as well as real ones, the encoding carries nothing and the project stops.

H1. Passes if agreement with the base model is below 95% and a majority of 30 blind-rated disagreements are judged substantive.

H3. Passes if the readout shift on linked contentions exceeds the shift on unrelated contentions by more than the hardware-versus-simulator noise floor, in the predicted direction, on at least two thirds of probed items.

H4. No threshold. Procedure: for every disagreement with the base model, a rater attempts to trace the layer's judgment back to a single input contention or to their weighted combination. Cases where neither trace succeeds are reported as candidate emergent judgments, each described in full, with the window's recording attached. The paper reports the count and the cases; it does not claim emergence from the count alone.

H5. Passes if at least one classic effect appears with the human sign.

Noise check. Hardware and simulator arms are compared throughout. If hardware results diverge from the simulator beyond a stated tolerance, hardware runs are reported as noise-limited and the simulator arm carries the primary analysis, stated plainly.

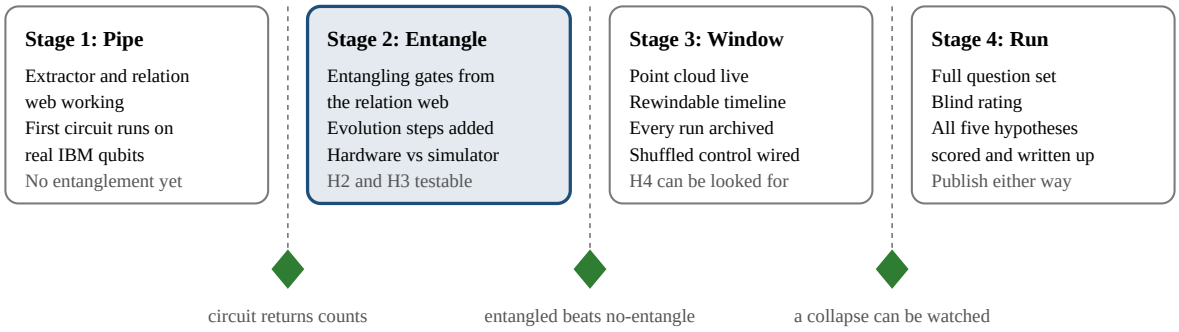
Stop rule. If H2 fails under both controls, the central claim is falsified and the write-up proceeds as a negative result.

Roadmap

The work is staged by result, not by date. Each stage ends when its gate is passed (Figure 8). Some of this can happen in a day; some will not; the sequence is what matters.

Figure 8

Four Stages, Each Gated by a Result



Note. No calendar is attached. A stage ends when its gate condition is met. Stage 2 is the stage on which the whole proposal rests.

Stage 1, the pipe. Extractor and relation judge calibrated on the belief pool. First circuit submitted to IBM hardware and counts returned. No entanglement yet; this proves the plumbing and nothing else. Gate: a real circuit returns real counts.

Stage 2, entangle. Entangling gates built from the relation web, evolution steps added, all four arms running. Hardware and simulator compared. Gate: the entangled arm beats the no-entanglement arm on a first batch of known-wrong items.

Stage 3, the window. Point cloud live, timeline scrubbable, every run archived, shuffled control wired in. Gate: a single collapse can be watched end to end and rewound.

Stage 4, run and write. Full 200-question set, all probes, blind rating, every hypothesis scored. Gate: results and conclusions sections filled, whatever they say.

Beyond Stage 4. If H2 holds, the layer is packaged as a service that any language model can call, and the archive of collapses becomes the asset a company is built on. If fault-tolerant hardware arrives on the timelines currently projected, the same architecture scales from sixteen contentions to thousands, and the question changes from whether a judgment can be entangled to whether a whole mind can.

Implications if the Hypotheses Hold

Physics. A demonstration that entanglement does useful cognitive work on today's noisy hardware would be a rare near-term quantum application that is not a speed-up claim. It would also give quantum computing a new kind of benchmark: not how fast, but how differently.

Psychology. Quantum cognition would move from a family of fitted models to a generative mechanism that can be built, perturbed, and observed. Every human effect it was designed to describe becomes something that can be induced in a machine and studied with full access to the state.

Philosophy of mind. The claim that a judging mind is a unity rather than a sum would acquire an experimental correlate. The hard problem would not be solved, but one of its load-bearing intuitions, that a mind is not decomposable into its beliefs, would have a working example outside a human skull. That is a different kind of argument than philosophy has had access to before.

Artificial intelligence. A judgment layer that lets contradictions cancel rather than average is a new component. It would sit between any model's reasoning and its answer, flag conflicts the model cannot see in itself, and expose the dynamics of a decision in a form a person can watch. That is both a safety tool and an interpretability tool, and neither exists today in this form.

Commercial. The layer is a service any model provider could call. The window is a product people would pay to look through. The archive of recorded collapses is a dataset that cannot be reconstructed from the outside and grows with every run. And the long bet, that when fault-tolerant hardware arrives the same architecture scales to a whole mind, is a position no incumbent currently holds.

Limitations and Scope

The encoding is a choice. Turning confidence into amplitude and relations into entangling gates is one mapping among many. The shuffled-relations control exists precisely to test whether this mapping carries information; it does not show the mapping is the right one.

Qubit budget. Free-tier hardware limits the layer to roughly sixteen contentions. Richer reasoning traces will be truncated or run on the simulator, and the simulator cannot show anything the hardware could not in principle, so hardware noise is a ceiling on what can be claimed.

Hardware noise. Today's devices are noisy enough that subtle interference effects may be washed out. The hardware-versus-simulator comparison is reported throughout, and any hypothesis that passes only on the simulator is marked as such.

Extraction error. Splitting a reasoning trace into contentions is itself a model judgment and will be imperfect. The first fifty are hand-checked; the error rate is reported.

Single rater. Blind rating of disagreements is done by one person in this first study. A second rater is the first thing to add in replication.

Model drift. The base model may change during the study. Model version is pinned and recorded.

Emergence is hard to pre-register honestly. H4 is given a procedure rather than a threshold, and the paper reports cases rather than a verdict. Readers should treat any emergence claim here as a pointer for further work, not a result.

Consciousness is out of scope. This work touches the hard problem by giving one of its intuitions an experimental handle. It makes no claim that the system is conscious, that the brain is quantum, or that entanglement is sufficient for mind.

Results

To be completed after Stage 4. Each hypothesis will be reported against its pre-registered criterion, with the hardware-versus-simulator comparison and the H4 case list.

Conclusions

To be completed after Results. Whatever the data say, the answer to the opening question, do machines need quantum minds, will be reported plainly, with the next experiment the result points to.

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